

too powerful, for the setups were also used.

It must also be clarified that the processors from AMD and Intel are not identical in performance, so it would be wrong to compare all the scores of the Intel boards with the AMD ones. Although the processor and motherboard were the only hardware differing, the two tests were handled separately.

Motherboards For Intel Processors

Intel is in a pretty comfortable position right now. Going by the number of people you come across, it appears like there are more Intel Core 2 Duo users these days than ever before. We received loads of products and we chose motherboards with the Intel onboard graphics chipsets – the G43's and the G45's. To keep up with chipset releases from all sides, Intel has been on a roll releasing updates for all of their chipsets before the whole Nehalem trend settles in, and the all new Socket LGA 1366 will have to be used. These might be the last few chipsets for the LGA 775 format it seems. The X48, P45, G45, G43 are the latest updates from Intel.

From the NVIDIA bandwagon, we received all the spanking new GeForce 9300 chipset-based motherboards. The ECS GF9300T-A, MSI P7NGM and the ZOTAC GeForce 9300 are the three boards using the new chipset.

Features

Features mean everything from components on the motherboards to things seen in the BIOS. The ASUS P5Q-EM is the best of all the boards when it comes down to just features. It comes with good BIOS with a lot that overclockers will like. Sensible increments and options to alter speeds and voltage for all the components is a God sent with this board. For a board that costs Rs 8,375, these are some of the features that are found in the high-end boards from ASUS and some other brands. Another board with similar overclocking features is the Gigabyte EG43M-S2H. The board also doesn't come with a PS2 port for the mouse, so you need to use a USB mouse or get a PS2 to USB convertor.

The Zotac GeForce 9300 board – although very good in performance, as you'll soon find out – is good, but the BIOS is rather boring. The BIOS is extremely basic with the absolute least amount of overclockability options and features. As for the features that are available, they just aren't practical. The huge voltage increments are an example of this. If you were planning on overclocking a lot, then this board is definitely a very bad idea. What it lacks in BIOS features, it makes up in the features on the board itself. The chip comes with a tiny fan on it, and this works well for the 9300. Typically, we've seen that the 9300 gets pretty hot when you're playing games.



Gigabyte EG43M-S2H is a complete board when it comes to features

Like some of the boards, VGA, DVI and even HDMI are available. The microATX form factor along with the various display connectivity options makes it a good HTPC board. The MSI P7NGM and the ECS GF9300T-A are also very similar with the number of features.

The MSI P7NGM like the ZOTAC too hardly has any overclocking features built into the BIOS. It looks very similar to the ZOTAC board, but this one doesn't come with a fan on top of the heat-sink; it uses solid state capacitors though. It has a clean design with the SATA ports positioned away from the PCI Express graphics card slot. The ECS GF9300T-A is no different from the other two 9300 boards either. The one thing that makes it stand out is the ALC889-based audio solution. The only other board with a different audio solution was the ASUS P5Q-EM with the Realtek ALC1200 chip. All the motherboards came with a single wired Gigabit LAN port. NVIDIA's HybridSLI and HybridPower allows external graphic cards to work better with the motherboard, both to save power as well as improve performance, but this works only for NVIDIA graphics cards.

In order to be future-proof, memory expandability is important, and only the ZOTAC GeForce 9300 and the Intel P5Q-EM can handle 16 GB of DDR2 so far. The MSI G43M2 board only comes with two RAM slots. None of the boards we received for the Intel platform supported DDR3 RAM or CrossFire, or even SLI.

Layout And Design

The ASUS P5Q-EM motherboard chipset cooler is bright blue, comes in an odd stealthy shape and does a good job of cooling the chip along with making the board look great. There is only one PS2 port that can be used for either a mouse or a keyboard. The board in general has been laidout very evenly and kept as spacious as possible even though this is a Micro ATX form factor board.

The ECS GF9300T-A is a proper ATX board and is the largest of all the motherboards for Intel processors. The SATA ports are placed above the PCI Express slot so that there is absolutely no way the card or the cooler can block the ports. In looks as well, the board is very gaudy and bright. It is a predominantly black board with a bright red PCI-E slot and black PCI slots. Other than the red and yellow RAM slots, there's also a fancy looking heatsink with a Black series badge on top. Dedicated power buttons on the board for Reset and Power On/Off are also a good touch for a board with a modest price. This is easily the least cramped board of all the ones for the Intel platform.

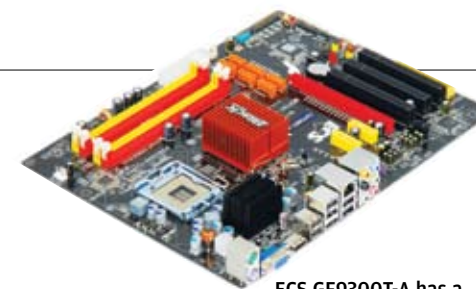
The Gigabyte EG43M-S2H on the other hand is a lot more cramped than the ECS. There are a few capacitors near the PCI Express slot that can be of concern when you plug in a large card such as a GeForce GTX 280 or a Radeon 4870 X2, for example. The heatsink on the north and south-bridge do a fair job of cooling. The MSI G43M2

has loads of capacitors placed all over the board between slots and even has the CMOS cell placed in between the PCI Express and the first PCI slot.

In the ZOTAC GeForce 9300, the 24-pin power connector for the board is located in an unusual position above the processor on the extreme edge. If you have happened to use a cabinet that has the power supply mounted at the bottom, then be sure the power supply has a long enough cable. There's also a LED poster with error codes if necessary. USB ports on the back of the board are also an unusual orange.

Performance

Performance is always an important factor no matter how many features a motherboard comes with. Going by overall performance, the Intel chipset motherboards were lagging behind the all new NVIDIA GeForce 9300 chipsets. Gaming performance is important in a test for motherboard with onboard graphics which is where the 9300s excelled.



ECS GF9300T-A has a clean spaced out layout

We also have to be clear that the Intel G43 and G45 chipsets have the better SATA controller than the NVIDIA ones. In all the data transfer tests where 4 GB files were copied to the drive, we noticed that the G43 and the G45 motherboards were three to four seconds faster. The MSI P7NGM board especially recorded rather poor scores in this test.

It was the same in the PC Mark tests. Although the 9300 boards performed a lot better in comparison to the Intel chipsets in the graphics tests of the benchmark suite, the real-world desktop application tests showed Intel chipsets doing better.

Of all the motherboards, the ZOTAC GeForce 9300 is a good performer, only slightly ahead of the ECS GF9300T-A motherboard. The only place where the ZOTAC GeForce 9300 lost out to the rest of the motherboards was in the file transfer tests. The 4 GB file transfer test took considerably longer than the other motherboards. Even the MSI P7NGM was lagging far behind the rest.

There was little or no difference in the CPU intensive video encoding tests. All of the boards scored between 67 and 70 seconds.

Pioneer And Leader In Technology

Radeon HD 4870 SONIC Dual Edition

Nickel-plated copper pipes

Dual Fans

Dual BIOS

Ultimate Dual Output

- DHMI(DVI)
- DISPLAY PORT
- DVI
- CRT

- GDDR5 512MB 256bit
- Overclocking Edition
- HDMI + DisplayPort + DVI + D-SUB
- Triple-Pipe cooler with dual fans
- Smart BIOS switch

Wanna know the latest products and technologies ?

Please visit Palit forum: <http://forum.palit.biz>

TIRUPATI ENTERPRISES
Bring You The Best Of Digital World
 VGA Distributor: TIRUPATI ENTERPRISES
 Tel: +91-33-22251192

Mediaman
 VGA Distributor: MEDIAMAN INFOTECH PVT. LTD.
 Tel: +91-22-23823100

aditya
 VGA Distributor: ADITYA INFOTECH LTD
 Tel: +91-11-25223810